AN LCI MORPHISM IS UNRAMIFIED IFF FORMALLY UNRAMIFIED, AND THEN ITS CONORMAL SHEAF IS FINITE LOCALLY FREE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 18708-18714 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 18716-18728, the complete author proof (11 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. First sentence of the proof is a pointer; the conormal computation is written out — retained as substantive.

Editorial note. Counted A side only (spaces twin proof incomplete per ruling v3).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. In-source lemmas unramified-formally-unramified and two-unramified-morphisms-formally-smooth; Divisors, lemma quasi-regular-immersion.

Source reference. Original lemma 063M: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18728–18728. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 067W: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3471–3490. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02HE: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 975–979. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a local complete intersection morphism of schemes. Then f is unramified if and only if f is formally unramified and in this case the conormal sheaf $\mathcal{C}_{X/Y}$ is finite locally free on X.*

Proof. The first assertion follows immediately from Lemma 02HE and the fact that a local complete intersection morphism is locally of finite type. To compute the conormal sheaf of f we choose, locally on X, a factorization of f as $f = p \circ i$ where $i : X \to V$ is a Koszul-regular immersion and $V \to Y$ is smooth. By Lemma

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067W we see that $\mathcal{C}_{X/Y}$ is a locally direct summand of $\mathcal{C}_{X/V}$ which is finite locally free as i is a Koszul-regular (hence quasi-regular) immersion, see Divisors, Lemma 063M. $\square$